Deok-Hyeon Cho

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Employment History

2022 – 2022 **RexSoft,** Intern,
Participated in debugging and development of the REX program in the Development Department.

2021 – 2021 **Visang Education,** Intern,
Assisted with data entry, validation, analysis, and report creation for the E-Learning Content Planning Department.

Education

2022 – · · · · **Korea University,** Seoul, South Korea.
Integrated M.S.&Ph.D. in Artificial Intelligence in Pattern Recognition & Machine Learning Lab, under the supervision of Seong-Whan Lee. (GPA: 4.19 / 4.50)

2016 – 2022 **Hanyang University,** Ansan, South Korea.
B.S. degree in Applied Mathematics. (GPA: 4.40 / 4.50)

Research Publications and Patents

Research Publications

1. **D.-H. Cho**, S.-B. Kim, H.-S. Oh, and S.-W. Lee, "Affectflow: Word-level affective flow modeling for conversational speech synthesis," in *(Under Review)*, 2026.
2. **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, "Affectron: Emotional speech synthesis with affective and contextually aligned nonverbal vocalizations," in *(Under Review)*, 2026.
3. H.-S. Oh, **D.-H. Cho**, S.-B. Kim, and S.-W. Lee, "Toward complex-valued neural networks for waveform generation," in *Proc. Conference of the International Conference on Learning Representations (ICLR)*, 2026.
4. **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, "Diemo-tts: Disentangled emotion representations via self-supervised distillation for cross-speaker emotion transfer in text-to-speech," in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2025.
5. **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, "Emosphere-ser: Enhancing speech emotion recognition through spherical representation with auxiliary classification," in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2025.
6. **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, "Emosphere++: Emotion-controllable zero-shot text-to-speech via emotion-adaptive spherical vector," in *IEEE Transactions on Affective Computing (TAFFC)*, 2025.
7. N.-G. Kim, **D.-H. Cho**, S.-B. Kim, and S.-W. Lee, "Spotlight-tts: Spotlighting the style via voiced-aware style extraction and style direction adjustment for expressive text-to-speech," in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2025.
8. H.-S. Oh, S.-H. Lee, **D.-H. Cho**, and S.-W. Lee, "Durflex-evc: Duration-flexible emotional voice conversion leveraging discrete representations without text alignment," in *IEEE Transactions on Affective Computing (TAFFC)*, 2025.

9. **D.-H. Cho**, H.-S. Oh, S.-B. Kim, S.-H. Lee, and S.-W. Lee, "Emosphere-tts: Emotional style and intensity modeling via spherical emotion vector for controllable emotional text-to-speech," in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2024.
10. J.-E. Lee, S.-B. Kim, **D.-H. Cho**, S.-H. Lee, and S.-W. Lee, "Promoticon: Prompt-based emotion controllable text-to-speech via prompt generation and matching," in *Proc. IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2024.

Domestic Patents

1. "Apparatus and method for speech synthesisn," *10-2025-0088028*, 2025.
2. "Method and system for expressive text-to-speech via voiced-aware style extraction and style direction adjustment," *10-2025-0116457*, 2025.
3. "Method for cross-speaker emotion transfer in text-to-speech using disentangled emotion representations via self-supervised distillation," *10-2025-0130127*, 2025.
4. "Emotional expression voice generation apparatus and method capable of controlling emotional style and intensity using continuous emotional dimensions," *10-2024-0029066*, 2024.
5. "Method, device, and program for synthesizing voices expressing emotions based on prompts," *10-2024-0099370*, 2024.

Miscellaneous Experience

Awards, Achievements, and Scholarship

2025 **Excellence Award in Extreme-Noise Speech Recognition & Restoration AI Model Development Competition**, Korea Artificial Intelligence Association (KAIA).

2022 **[BK21 Fellowship] Securing Outstanding Graduate Students**, Korea University.

Department Prize for Outstanding Student Performance, Hanyang University.

2021 **Excellence Award in the Credit Card User Delinquency Prediction AI Competition,** organized by Hanyang University on Dacon.

Reviewer

2024- IEEE International Conference on Systems, Man, and Cybernetics (SMC)

2025- IEEE International Conference on Acoustics Speech and Signal Processing (ICASSP)

IEEE Transactions on Affective Computing (TAFFC)

IEEE Signal Processing Letters (SPL)

2026- International Conference of the International Speech Communication Association (INTERSPEECH)

Research Projects

2024 - 2024 **Samsung Research**, South Korea.
Development of interjection utterances for natural speech synthesis.

2025 - 2025 **Thomas Crown**, USA.
Voltron: Cross-speaker Controllable Emotion Transfer TTS.

2025 - 2026 **Murf AI**, USA.
Full Duplex Speech-to-Speech Conversational AI.